

2.4. Radiomic Feature Selection

A hierarchical four-step feature selection framework was implemented to identify robust and biologically meaningful radiomic features while minimizing overfitting and information leakage. All feature selection procedures were embedded within the training data of each outer fold of the nested CV framework. Consequently, the corresponding held-out test fold remained completely unseen during feature selection, preprocessing, hyperparameter optimization, and model training.

Stage 1: Volume-Independence Filtering

In the first stage, volume-independent filtering (VIF) was applied to reduce the influence of lesion size on the radiomic signature. Within each outer training fold, Spearman rank correlation coefficients were calculated between the predefined lesion volume feature and each candidate radiomic feature. Features demonstrating an absolute Spearman correlation greater than 0.50 (absolute Spearman $|r| \geq 0.50$) with lesion volume were considered volume-dependent and excluded from further analysis. Morphological features were flagged for reporting purposes, but no feature category was removed solely based on its name or class. To account for multiple comparisons, p-values were adjusted using the Benjamini–Hochberg false discovery rate (FDR) procedure and only features with q-values < 0.01 were retained. This initial screening step reduced dimensionality by eliminating features lacking significant associations with the target endpoint.

Stage 2: Redundancy Reduction

Subsequently, redundancy reduction was conducted using pairwise Pearson correlation analysis. Highly correlated features ($|r| > 0.90$) were considered redundant and grouped accordingly. Within each correlated group, features were ranked by their discriminative performance, quantified by the area under the receiver operating characteristic curve (AUC). The highest-ranking feature was retained, whereas the remaining correlated features were removed. This approach reduced redundancy while preserving the most informative representative variables.

Stage 3: Minimum Redundancy Maximum Relevance (mRMR) Selection

Following redundancy and multicollinearity reduction, minimum redundancy maximum relevance (mRMR) feature selection was applied in a modality-balanced manner. PET and CT features were first separated into independent feature blocks. Within each modality, mRMR prioritized features that maximized relevance to the target outcome while minimizing redundancy with already selected features. Relevance was quantified using mutual information, whereas redundancy was estimated from pairwise feature correlations. To ensure balanced representation of both imaging modalities, a greedy forward selection procedure was used to retain a maximum of five features from each modality before combining the selected PET and CT feature subsets into a unified radiomic signature.

Stage 4: Stability Selection

As the final feature selection stage, feature robustness and reproducibility were assessed using Elastic-Net stability selection. Candidate features were evaluated across 2,000 stratified subsampling iterations, each using 60% of the available training samples. Within each iteration, Elastic-Net logistic regression models were fitted using predefined L1-ratio values of 0.60, 0.70, 0.80, 0.90, and 0.95, combined with inverse regularization strengths (C) of 0.001, 0.01, 0.05, 0.10, and 0.50. Features with absolute regression coefficients exceeding 10^{-3} were considered selected in a given iteration. Selection frequency was subsequently calculated as the proportion of iterations in which a feature was selected, and features with selection frequencies ≥ 0.75 were retained in the final radiomic signature. To avoid overly restrictive selection resulting in an insufficient number of predictors, the selection frequency threshold was iteratively reduced in 0.05 increments to a minimum of 0.55 whenever fewer than two features satisfied the primary criterion. If fewer than two features remained after threshold relaxation, the two most frequently selected features were retained to ensure a minimum feature set for subsequent model development.

The resulting feature selection framework integrated statistical significance assessment, volume-independence filtering, redundancy reduction, multicollinearity control, modality-balanced relevance optimization, and stability selection. This hierarchical strategy yielded a compact set of informative, non-redundant, and reproducible radiomic biomarkers for subsequent machine learning analysis.

2.5. Preprocessing and Machine Learning Model Construction

The dataset exhibited moderate class imbalance between physiological and pathological uptake regions. To address this issue while avoiding information leakage, the Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively within the training partition of each outer CV fold [29]. Synthetic minority-class samples were generated solely from the training data, whereas validation and test partitions remained untouched throughout the modeling process. This strategy preserved the original class distribution in the evaluation data and enabled unbiased estimation of model performance under clinically realistic conditions. Given the limited cohort size, SMOTE was employed strictly as a training-time augmentation strategy to improve classifier learning and decision boundary estimation rather than to alter the underlying prevalence of the target classes.

To investigate the impact of imaging modality and gray-level discretization on predictive performance, four independent radiomics experiments were conducted. The first two experiments utilized PET radiomic features alone, extracted using discretization settings corresponding to 64 and 128 gray levels, respectively. The remaining two experiments employed combined PET/CT radiomic features, using the same PET discretization settings (64 and 128 gray levels) while incorporating CT-derived radiomic features. Consequently, the evaluated configurations consisted of: (i) PET-only (64 gray levels), (ii) PET-only (128 gray levels), (iii) PET/CT (64 gray levels), and (iv) PET/CT (128 gray levels). Each experimental setting underwent an identical feature selection, model optimization, and validation

pipeline to enable a fair comparison of modality-specific and discretization-dependent effects on classification performance.

Model development and performance assessment were conducted using a fully nested CV framework designed to provide unbiased estimates of generalization performance while preventing information leakage during feature selection and hyperparameter optimization. The outer loop employed a 4-fold StratifiedGroupKFold scheme for model evaluation, whereas the inner loop utilized a 3-fold StratifiedGroupKFold strategy for hyperparameter optimization. The selection of four outer folds represented a practical compromise between maintaining adequate training data for model development and ensuring sufficient independent test samples for reliable performance estimation, given the limited cohort size (24 patients and 734 segmented regions).

To ensure strict patient-level independence, all regions originating from a given patient were assigned exclusively to either the training or testing partition within each fold. Consequently, no patient contributed samples to both sets simultaneously. This group-aware validation strategy prevented overly optimistic performance estimates arising from intra-patient correlations and ensured that model evaluation reflected the clinically relevant scenario of predicting outcomes in previously unseen patients.

Prior to model training, radiomic features were standardized using z-score normalization. Scaling parameters were estimated exclusively from the training data within each outer fold and subsequently applied to the corresponding validation and test sets. All preprocessing operations, including feature selection, scaling, and class balancing, were performed entirely within the training data of each CV iteration to eliminate potential information leakage.

Hyperparameter optimization was performed using Optuna (v3.5.0) with Tree-structured Parzen Estimator (TPE) sampling. For each candidate model, 30 optimization trials were conducted within the inner CV loop, with the mean receiver operating characteristic area under the curve (ROC-AUC) serving as the optimization objective. Search spaces were defined to encompass model-specific parameters, including regularization strength for linear and kernel-based methods, tree depth and ensemble size for tree-based learners, learning rates for boosting algorithms, and neighborhood size for instance-based classifiers.

To comprehensively evaluate the predictive value of the radiomic signature across diverse learning paradigms, ten supervised machine learning algorithms were investigated: Logistic Regression, Linear Support Vector Machine (Linear SVM), Radial Basis Function Support Vector Machine (RBF-SVM), Random Forest, Extra Trees, Gradient Boosting Machine, XGBoost, LightGBM, K-Nearest Neighbors (KNN), and Gaussian Naïve Bayes. These models collectively represent linear, kernel-based, ensemble, boosting, instance-based, and probabilistic learning approaches. All models were implemented using scikit-learn (v1.3.0), XGBoost (v2.0.3), and LightGBM (v4.1.0), with fixed random seeds to ensure reproducibility of all analyses.

2.6. Machine Learning Model Performance Evaluation and Explainability

Model performance was comprehensively evaluated using multiple complementary metrics to assess discrimination, classification accuracy, calibration, and potential clinical utility. Performance measures included the area under the receiver operating characteristic curve (ROC-AUC), precision–recall area under the curve (PR-AUC), accuracy, balanced accuracy, precision, recall (sensitivity), specificity, and F1-score. Balanced accuracy was included to account for class imbalance and to provide a more reliable estimate of classifier performance across both classes. In addition, confusion matrices were generated to quantify true-positive, true-negative, false-positive, and false-negative predictions, thereby enabling detailed assessment of classification behavior.

To statistically compare classifier performance, pairwise DeLong tests were performed using out-of-fold predictions obtained exclusively from the independent test partitions of each outer CV fold. Predictions were aggregated only after completion of the entire CV procedure, ensuring strict separation between training and testing data and preventing information leakage. To account for multiple pairwise comparisons among the eleven evaluated classifiers (55 comparisons), Bonferroni correction was applied, resulting in a corrected significance threshold of $\alpha = 0.00091$.

Model interpretability was investigated using Shapley Additive Explanations (SHAP) [30]. SHAP values are derived from cooperative game theory and quantify the contribution of each feature to an individual prediction by measuring its average marginal effect across all possible feature combinations. For a given feature and input instance, the SHAP value represents the contribution of that feature to the difference between the model prediction and the baseline prediction. Because the exact computation of Shapley values becomes computationally intractable for high-dimensional radiomics data, model-specific approximation methods were employed. SHAP values were calculated for each test instance, enabling both local interpretation at the individual region level and global interpretation across the entire cohort.

To obtain robust estimates of feature importance, SHAP values were aggregated across all outer CV test folds. Positive SHAP values indicated an increased likelihood of classification as pathological uptake, whereas negative values favored physiological uptake. This analysis was used to quantify and rank the relative contributions of PET- and CT-derived radiomic features, identify the imaging characteristics driving model predictions, and assess whether the learned decision patterns were consistent with established biological and radiological knowledge of physiological and pathological [^{68}Ga]Ga-FAPI-46 uptake. The primary objective of the SHAP analysis was to improve model transparency and facilitate interpretation of the derived radiomic signatures rather than to provide autonomous clinical decision support.

To further explore the structure of the radiomics feature space, t-distributed Stochastic Neighbor Embedding (t-SNE) was applied as a nonlinear dimensionality reduction technique. High-dimensional radiomic data were projected into a lower-dimensional representation while preserving local similarity

relationships between samples. The resulting visualizations enabled qualitative assessment of class separability and provided additional insight into the discriminative capability of the extracted radiomic features.

Calibration analysis was performed to evaluate the agreement between predicted probabilities and observed outcomes. Calibration curves were generated by comparing model-estimated probabilities with empirical event frequencies across probability intervals. A perfectly calibrated model corresponds to the diagonal reference line, whereas deviations from this line indicate systematic overestimation or underestimation of risk. This analysis provided insight into the reliability of probabilistic predictions and their potential suitability for clinical decision support.

Finally, decision curve analysis (DCA) was conducted to assess the potential clinical utility of the developed models across a range of decision thresholds. DCA quantifies the net benefit of a predictive model by jointly considering the consequences of true-positive and false-positive classifications and comparing model-guided decision-making with default strategies such as treating all cases or treating none. This approach enabled evaluation of the practical clinical value of the radiomics-based machine learning models for differentiating physiological from pathological uptake on [⁶⁸Ga]Ga-FAPI-46 PET/CT images.

3. RESULTS

Table 2 presents the mean ROC-AUC values and standard deviations for ten ML models evaluated across PET-only and PET/CT radiomics datasets using 64- and 128-bin grey-level discretization. In PET-only analyses, the 64-bin setting generally yielded higher ROC-AUC values than the 128-bin setting, particularly for tree-based ensemble models. Extra Trees, Random Forest, Gradient Boosting, and LightGBM achieved mean ROC-AUC values of 0.91 with 64-bin discretization, compared with 0.87–0.88 with 128-bin discretization. XGBoost showed a similar pattern, with a mean ROC-AUC of 0.90 using 64 bins and 0.87 using 128 bins.

The addition of CT-derived radiomic features improved discrimination for the strongest-performing ensemble models. In the PET/CT datasets, Random Forest, Gradient Boosting, LightGBM, and XGBoost achieved mean ROC-AUC values of 0.93–0.94, exceeding their corresponding PET-only performance. This improvement was most evident for PET/CT models using 64-bin discretization, for which Gradient Boosting, LightGBM, and XGBoost reached mean ROC-AUC values of 0.936, 0.937, and 0.94, respectively. By contrast, simpler classifiers, including logistic regression, linear SVM, Naive Bayes, and SVM-RBF, did not show comparable gains after the addition of CT features. Overall, the results indicate that the added value of CT was model-dependent and was primarily observed among tree-based ensemble classifiers, whereas the effect of grey-level discretization differed between PET-only and PET/CT feature sets.

Table 2: Mean ROC-AUC values $\pm$ standard deviation for 11 machine-learning models across four datasets and two grey-level discretization settings (64 and 128 bins). Higher values indicate better classification performance.

Model	PET-only (bin128)	PET-only (bin64)	PET/CT (bin64)	PET/CT (bin128)
Naïve Bayes	0.85 $\pm$ 0.05	0.88 $\pm$ 0.04	0.797 $\pm$ 0.033	0.81 $\pm$ 0.01
Logistic Regression	0.84 $\pm$ 0.03	0.81 $\pm$ 0.04	0.752 $\pm$ 0.032	0.77 $\pm$ 0.03
Extra Trees	0.87 $\pm$ 0.03	0.91 $\pm$ 0.03	0.904 $\pm$ 0.015	0.90 $\pm$ 0.04
SVM Linear	0.84 $\pm$ 0.03	0.81 $\pm$ 0.04	0.748 $\pm$ 0.032	0.76 $\pm$ 0.03
SVM_RBF	0.86 $\pm$ 0.04	0.89 $\pm$ 0.03	0.828 $\pm$ 0.027	0.83 $\pm$ 0.02
KNN	0.85 $\pm$ 0.04	0.88 $\pm$ 0.03	0.863 $\pm$ 0.028	0.85 $\pm$ 0.03
Random Forest	0.87 $\pm$ 0.04	0.91 $\pm$ 0.01	0.930 $\pm$ 0.021	0.93 $\pm$ 0.03
Gradient Boosting	0.88 $\pm$ 0.05	0.91 $\pm$ 0.01	0.936 $\pm$ 0.020	0.93 $\pm$ 0.03
Light GBM	0.88 $\pm$ 0.04	0.91 $\pm$ 0.01	0.937 $\pm$ 0.018	0.93 $\pm$ 0.03
XGBoost	0.87 $\pm$ 0.05	0.90 $\pm$ 0.01	0.94 $\pm$ 0.021	0.93 $\pm$ 0.03

Four experimental settings were compared based on image type (PET-only vs. PET/CT) and gray-level discretization (bin64 vs. bin128). The best overall performance was achieved in the PET/CT with bin64 setting, which is summarized in Table 3. Results for the remaining three settings (PET-only bin64, PET-only bin128, and PET/CT bin128) are provided in Supplementary Tables S1–S3, respectively.

Table 3 summarizes the performance of ten ML models on the PET/CT dataset. XGBoost and GradientBoosting achieved the highest AUC (0.936 $\pm$ 0.033), PR-AUC (0.961), and F1-score ($\sim$ 0.876). XGBoost leads in accuracy (0.849), balanced accuracy (0.855), precision (0.915), and specificity (0.863). RandomForest and LightGBM follow closely, while linear models (Logistic Regression, Linear SVM) and NaiveBayes perform worst. Ensemble tree-based methods consistently outperform all other classifiers.

Compared with the other three experimental settings (see Supplementary Tables S1–S3), the PET/CT bin64 configuration consistently produced superior discriminative performance, particularly for ensemble methods. This indicates that combining PET and CT information with a moderate gray-level binning (64 levels) provides the most favorable radiomic feature representation for the current classification task.

Table 3. Performance metrics (mean $\pm$ SD where applicable) for all machine learning models in the best-performing experimental setting: PET/CT with 64 gray levels (bin64).

Model	AUC	PR-AUC	Accuracy	Balance d Accurac y	Precisio n	Recall	F1	Specificit y
XGBoost	0.948 ± 0.024	0.96 6 ± 0.01 7	0.866 ± 0.052	0.869 ± 0.044	0.909 ± 0.036	0.868 ± 0.076	0.88 6 ± 0.04 7	0.870 ± 0.026
Random Forest	0.946 ± 0.028	0.96 5 ± 0.01 9	0.850 ± 0.049	0.864 ± 0.038	0.919 ± 0.040	0.835 ± 0.088	0.87 1 ± 0.04 3	0.893 ± 0.039
Gradient Boosting	0.945 ± 0.025	0.96 4 ± 0.01 8	0.849 ± 0.044	0.864 ± 0.035	0.923 ± 0.040	0.827 ± 0.081	0.86 9 ± 0.04 3	0.901 ± 0.035
Light GBM	0.941 ± 0.021	0.96 2 ± 0.01 7	0.834 ± 0.045	0.844 ± 0.041	0.896 ± 0.056	0.822 ± 0.076	0.85 5 ± 0.04 9	0.865 ± 0.048
Extra Trees	0.921 ± 0.031	0.94 5 ± 0.03 4	0.842 ± 0.056	0.825 ± 0.039	0.870 ± 0.041	0.855 ± 0.097	0.86 1 ± 0.06 7	0.794 ± 0.035
KNN	0.885 ± 0.049	0.89 1 ± 0.06 1	0.836 ± 0.060	0.835 ± 0.065	0.863 ± 0.082	0.860 ± 0.055	0.86 0 ± 0.06 1	0.810 ± 0.086
SVM (RBF)	0.857 ± 0.039	0.86 9 ± 0.03 8	0.800 ± 0.070	0.784 ± 0.054	0.834 ± 0.065	0.825 ± 0.095	0.82 8 ± 0.07 6	0.742 ± 0.031
Naïve Bayes	0.816 ± 0.070	0.83 7 ± 0.08 2	0.768 ± 0.050	0.739 ± 0.061	0.757 ± 0.114	0.940 ± 0.051	0.83 1 ± 0.04 4	0.538 ± 0.165
SVM (Linear)	0.741 ± 0.029	0.75 9 ± 0.03 7	0.735 ± 0.069	0.680 ± 0.030	0.748 ± 0.070	0.814 ± 0.174	0.77 3 ± 0.115	0.546 ± 0.121
Logistic Regression	0.740 ± 0.029	0.75 5 ± 0.03 8	0.722 ± 0.075	0.692 ± 0.040	0.788 ± 0.050	0.712 ± 0.160	0.74 3 ± 0.110	0.671 ± 0.124

Figure 2 presents a panel of ROC curves comparing the discriminative performance of all ten machine learning models across the four experimental settings defined by image type (PET-only vs. PET/CT) and gray-level discretization (bin64 vs. bin128). Each subplot corresponds to one setting, with the false-positive rate on the X-axis and the true-positive rate on the Y-axis.

Consistent with the quantitative findings reported in Table 3 (PET/CT bin64), the ROC curves in the corresponding subplot show the highest and most leftward-upward concentration, particularly for ensemble models including XGBoost, Gradient Boosting, Light GBM, and Random Forest. For the PET-only settings (bin64 and bin128), the ROC curves are visibly shifted toward the diagonal, indicating lower overall separability. The PET/CT bin128 setting yields curves that are intermediate between PET-only and PET/CT bin64, suggesting that increasing the gray-level range to 128 does not further improve and may slightly degrade model performance when both modalities are used.

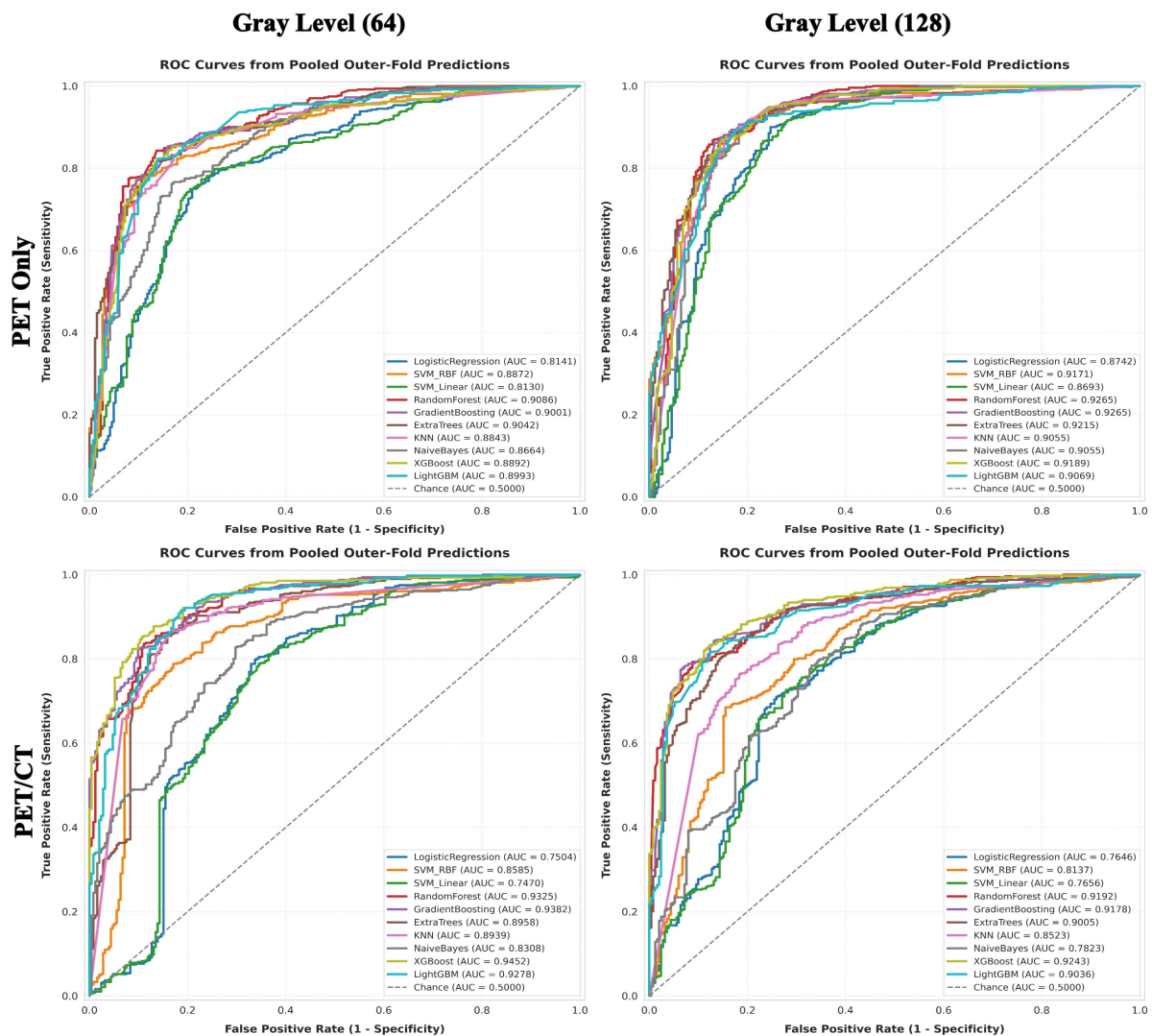


Figure 2. Panel of ROC curves for all ten machine learning models across four experimental settings: (A) PET-only bin64, (B) PET-only bin128, (C) PET/CT bin64, and (D) PET/CT bin128. The X-axis represents the false-positive rate ($1 - \text{specificity}$), and the Y-axis represents the true-positive rate (sensitivity).

Figure 3 displays decision curves for all ten machine learning models evaluated on the PET/CT bin64 dataset, plotting net benefit as a function of threshold probability. Reference lines for "Treat All" and "Treat None" strategies are included as benchmarks.

Across threshold probabilities ranging from approximately 0.1 to 0.7, XGBoost, LightGBM, and GradientBoosting consistently demonstrate the highest net benefit, indicating superior clinical utility. Among these, XGBoost maintains the greatest net benefit across the widest range of thresholds, particularly between 0.2 and 0.6. RandomForest and ExtraTrees show intermediate performance, with net benefit values slightly below the top ensemble methods but still exceeding both reference strategies for most thresholds.

Logistic Regression, SVM-RBF, and Linear SVM produce net benefit curves that overlap closely with or fall below the "Treat None" line beyond threshold probabilities of approximately 0.4, suggesting limited clinical applicability. Naive Bayes and KNN demonstrate moderate net benefit at low thresholds but decline rapidly as the threshold increases.

Overall, the decision curve analysis reinforces the quantitative findings from Table 3, confirming that ensemble tree-based models, particularly XGBoost, offer not only high discriminative performance but also meaningful clinical net benefit across clinically relevant threshold ranges.

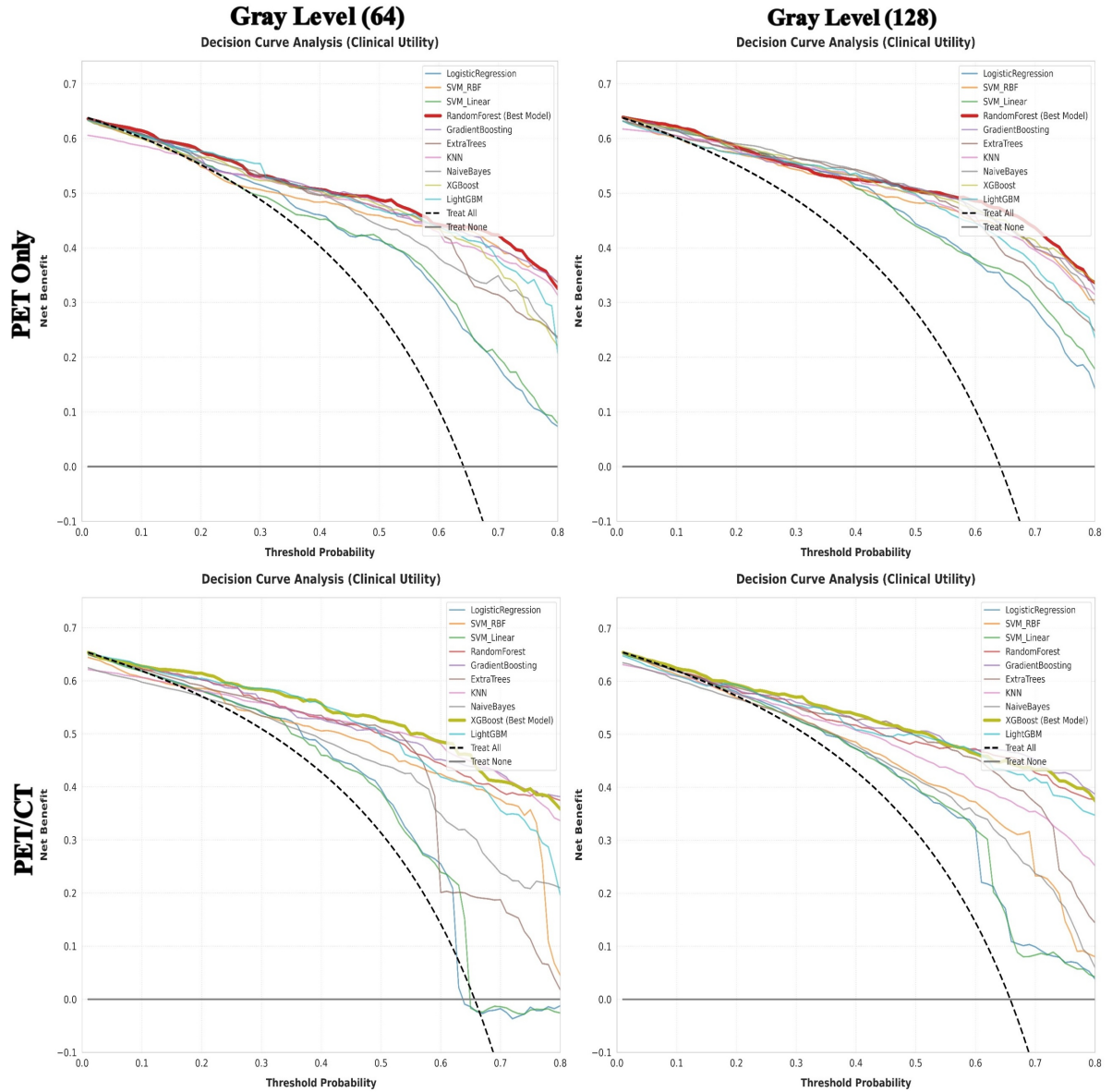


Figure 3. Decision curve analysis (DCA) for all ten machine learning models on the PET/CT bin64 dataset. The X-axis represents the threshold probability, and the Y-axis represents the net benefit. Reference lines for "Treat All" and "Treat None" strategies are shown.

Figure 4 presents a pairwise DeLong test comparison of AUCs for all ten models on the PET/CT bin64 dataset. The figure is organized into two complementary panels.

The left side (or lower triangle) displays the statistical significance of AUC differences as $-\log_{10}(p)$ -values. Values above approximately 1.3 correspond to $p < 0.05$, while values above 2 indicate $p < 0.01$. Ensemble methods, particularly XGBoost, GradientBoosting, and LightGBM, show high $-\log_{10}(p)$ values (frequently > 4) when compared against linear models (Logistic Regression, Linear SVM) and Naive Bayes, indicating highly significant AUC superiority. In contrast, comparisons

among the top three ensemble models yield low $-\log_{10}(p)$ values (< 1.3), suggesting no statistically significant difference between them.

The right side (or upper triangle) displays the AUC difference (row AUC minus column AUC), color-coded such that red cells indicate the row model has a higher AUC (row better). Blue cells indicate the column model has a higher AUC (column better). Consistent with the significance panel, red cells dominate when comparing ensemble models against linear or probabilistic classifiers, confirming the numerical superiority of tree-based methods. Blue cells appear primarily in comparisons where linear models or Naive Bayes are the row model and an ensemble method is the column model. Among the top ensemble models (XGBoost, GradientBoosting, LightGBM), the AUC differences are small (typically < 0.01), and the color is near-neutral, reflecting their closely matched performance.

Taken together, the two sides of Figure 4 provide both statistical and practical interpretations: ensemble methods, especially XGBoost, GradientBoosting, and LightGBM, achieve numerically higher and statistically significant AUC advantages over simpler models, while performing comparably to one another.

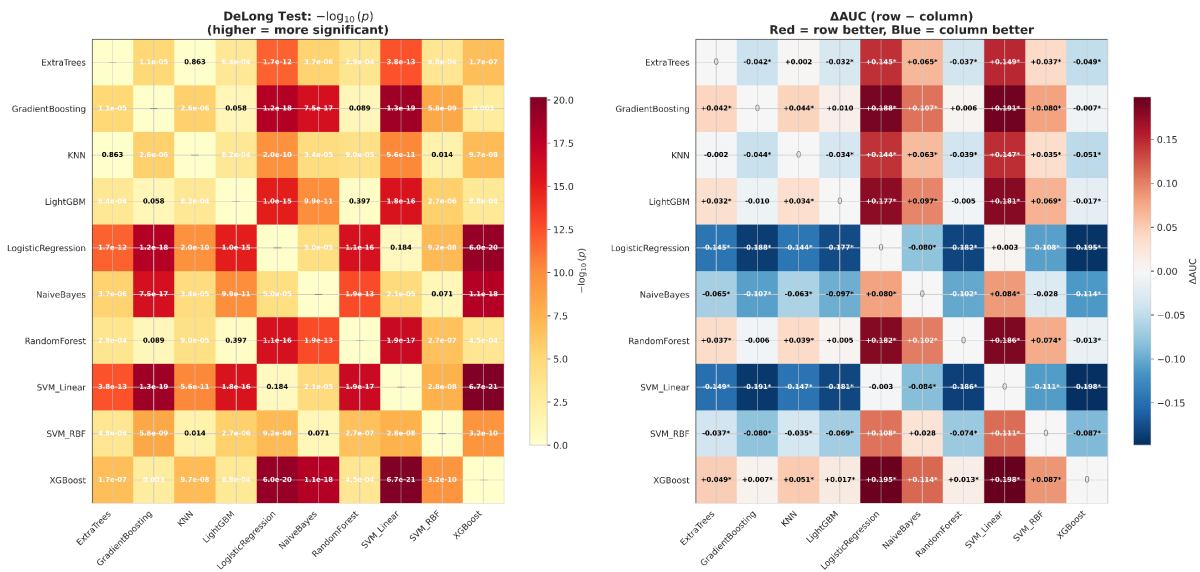


Figure 4. Pairwise DeLong test comparison of AUCs for all ten models on the PET/CT bin64 dataset. Left side (or lower triangle): $-\log_{10}(p)$ -values, where higher values indicate more significant differences (thresholds: >1.3 for $p < 0.05$, >2 for $p < 0.01$). Right side (or upper triangle): AUC difference (row AUC minus column AUC), with red indicating the row model is better and blue indicating the column model is better.

Figure 5 presents confusion matrices for the best-performing models derived from PET and PET/CT images at gray-level quantization levels of 64 and 128 bins. For PET alone at 64 bins, the model yielded 223 true negatives (TN), 40 false positives (FP), 73 false negatives (FN), and 398 true

positives (TP). Increasing to 128 bins markedly improved TP to 413 but drastically reduced TN to 221, indicating a loss of specificity. In contrast, PET/CT at 64 bins produced 219 TN, 33 FP, 64 FN, and 418 TP, while at 128 bins, it achieved 217 TN, 34 FP, 79 FN, and 404 TP. Thus, PET/CT demonstrated more stable performance across bin levels compared to PET alone, maintaining a consistent TN while substantially increasing TP. The best overall performance was observed for PET/CT with 64 bins, which delivered the highest TP (418) and a balanced error profile with only 33 FP and 64 FN, making it the optimal configuration for this classification task.

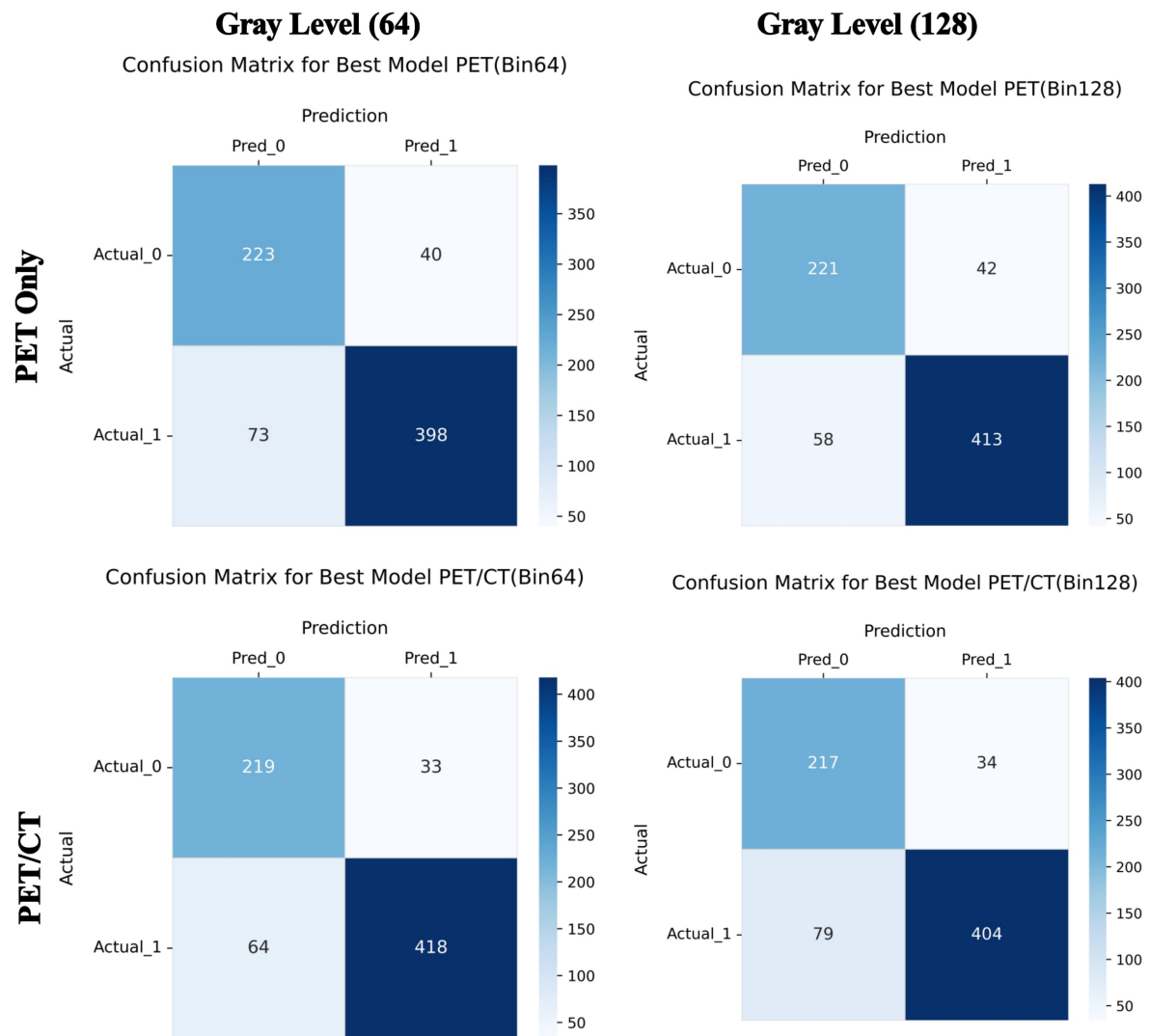


Figure 5. Confusion matrices for the bestperforming models using PET and PET/CT images at two graylevel quantization levels (64 and 128 bins). Numbers represent case counts for binary classification (Pred_0 = negative, Pred_1 = positive).

Figure 6 displays calibration curves (quantile binning) with Brier scores for ten classifiers across four evaluation settings. The diagonal represents perfect calibration, where predicted probabilities match observed frequencies.

Several consistent patterns emerge. Treebased ensembles, particularly *Random Forest* and *XGBoost*, achieve the lowest Brier scores across nearly all gray-level settings. At 128 gray levels (Setting 2), Random Forest yields the best calibration (Brier = 0.1021); at 64 gray levels (Setting 3), XGBoost is superior (Brier = 0.0983). Their curves closely follow the perfect calibration line, indicating minimal bias in predicted probabilities regardless of PET quantization depth.

Gradient Boosting and Light GBM perform competitively but with slightly higher Brier values (e.g., 0.1063 and 0.1179 at 128 gray levels, Setting 2). Extra Trees and KNN show moderate calibration, though Extra Trees exhibits wider variability (Brier from 0.1142 to 0.1564) across gray level settings.

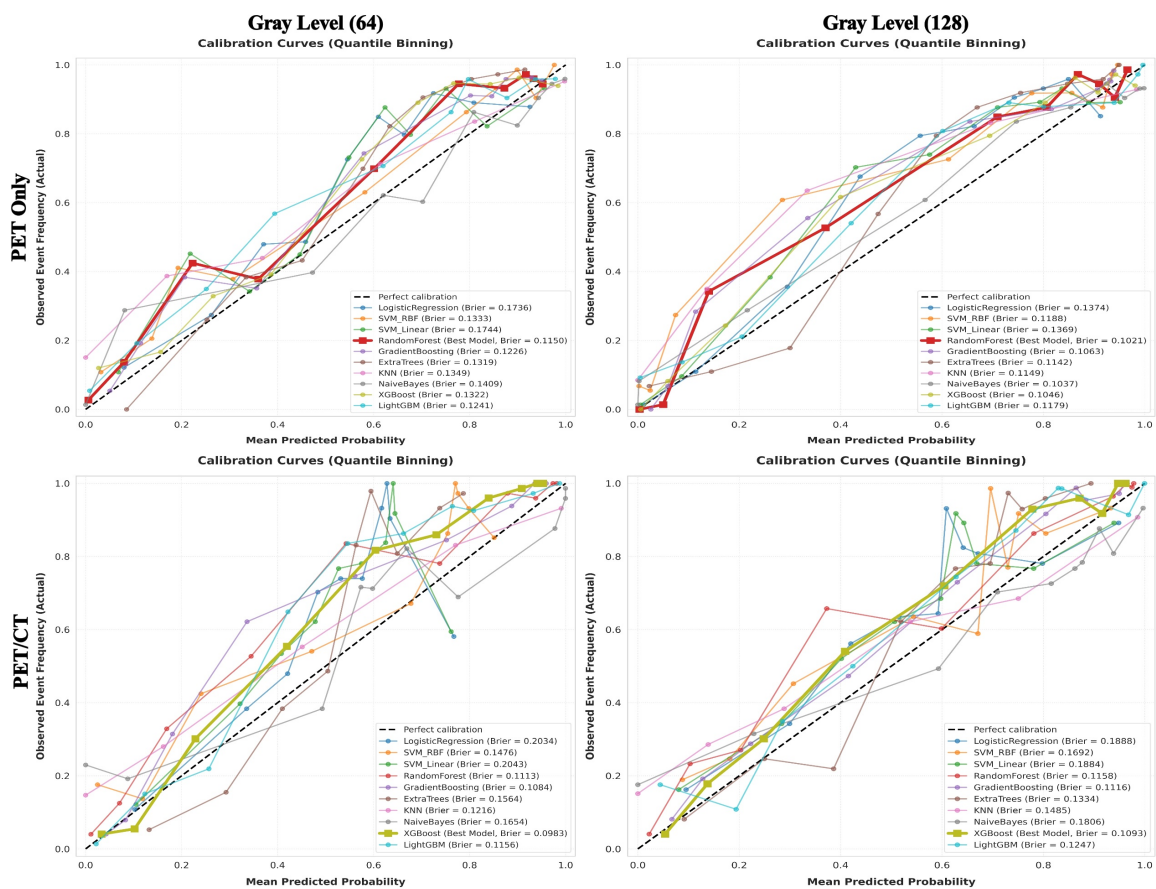


Figure 6. Calibration curves for all classifiers across PET gray level settings. Quantile binning calibration curves showing predicted probability (x-axis) versus observed frequency (y-axis) for ten classifiers. The dashed diagonal line represents perfect calibration. Each panel corresponds to one of four experimental settings based on PET image quantization: 64 gray levels (Settings 3 and 4) and 128 gray levels (Settings 1 and 2), with two distinct train/test configurations per gray level. Brier scores (reported in parentheses) quantify calibration error, where lower values indicate better calibration.

Figure 7 presents the two-dimensional t-distributed Stochastic Neighbor Embedding (t-SNE) projections of lesion feature embeddings generated under two gray-level discretization settings (64 and 128 gray levels) for both PET-only and PET/CT-derived radiomic features. Each point represents an individual lesion, while colors denote different patients.

The t-SNE maps reveal distinct differences in the organization of lesion feature space across imaging modalities and gray-level quantization schemes. In the PET-only analyses (top row), lesions form several compact and well-defined clusters, indicating the presence of coherent radiomic phenotypes. Notably, lesions from different patients are intermingled within many of these clusters, suggesting that the extracted PET radiomic features capture lesion-specific characteristics that are not solely driven by patient identity. This pattern is observed for both gray-level settings, although the 128 gray-level discretization appears to produce slightly smoother cluster transitions and a more continuous embedding structure.

In contrast, the PET/CT feature embeddings (bottom row) exhibit a more dispersed distribution with less clearly separated clusters. While localized groupings remain visible, the overall feature space appears more heterogeneous, reflecting the increased complexity introduced by combining metabolic (PET) and anatomical (CT) information. Several regions demonstrate partial patient-specific aggregation, particularly among patients with larger lesion burdens, indicating that multimodal radiomic signatures may capture subject-level imaging characteristics in addition to lesion-level heterogeneity.

Comparing gray-level discretization schemes, the global topology of the embeddings remains largely preserved between 64 and 128 gray levels, suggesting that the radiomic representation is relatively robust to gray-level quantization. However, the higher gray-level setting yields slightly enhanced local neighborhood structure, particularly within PET-only embeddings, where clusters appear more compact and internally coherent. This observation indicates that finer intensity discretization may improve the preservation of subtle textural differences between lesions without substantially altering the overall feature landscape.

Overall, the t-SNE analysis demonstrates that radiomic features generate meaningful low-dimensional representations of lesion heterogeneity, with PET-derived features exhibiting stronger intrinsic clustering than the combined PET/CT feature set. The consistency of cluster organization across gray-level settings further supports the stability of the extracted radiomic signatures.

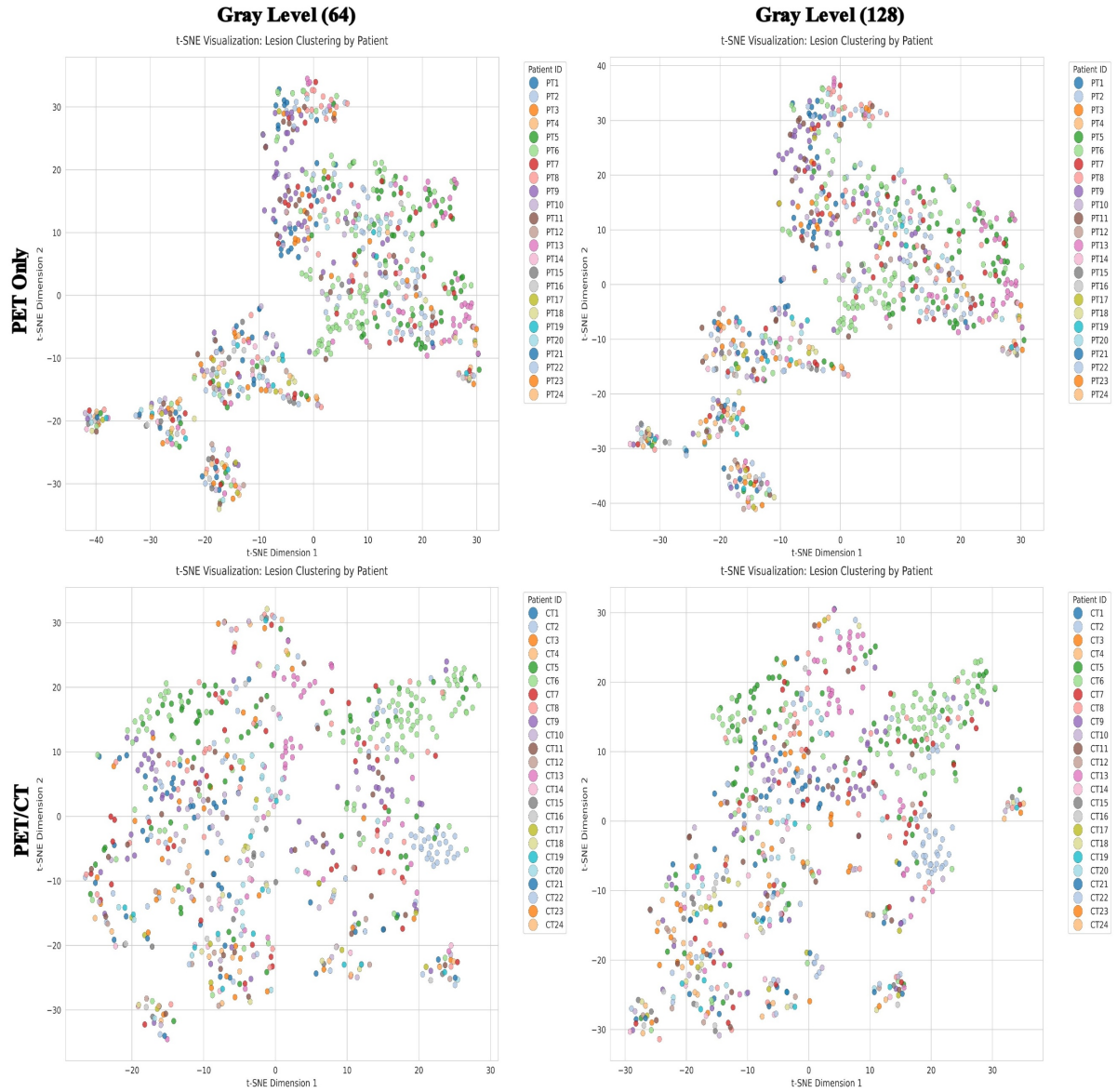


Figure 7. Two-dimensional t-distributed Stochastic Neighbor Embedding (t-SNE) visualization of lesion feature embeddings stratified by patient and gray-level quantization settings.

Figure 8 illustrates the two-dimensional t-SNE projections of lesion radiomic feature embeddings colored according to the target classification labels (Class 0 and Class 1) for both PET-only and combined PET/CT feature sets under gray-level discretization settings of 64 and 128. The visualization was used to assess the intrinsic separability of the classes within the radiomic feature space before classification modeling.

For the PET-only feature sets (top row), a clear distinction between the two target classes is evident. Lesions belonging to Class 0 predominantly occupy the lower and left regions of the embedding space, whereas Class 1 lesions are concentrated in the central and upper regions. Although a limited degree of overlap exists near cluster boundaries, the overall separation remains pronounced across both gray-level settings. This distinct class organization suggests that PET-derived radiomic features

capture biologically relevant patterns associated with the classification outcome and possess strong discriminatory capability.

The 128 gray-level PET embedding demonstrates a slightly more structured separation than the 64 gray-level counterpart, with clearer cluster boundaries and reduced mixing between classes. This observation indicates that finer gray-level discretization may enhance the representation of subtle textural characteristics relevant to class differentiation.

For the combined PET/CT feature embeddings (bottom row), class separation remains visible but is less distinct than in the PET-only analysis. The two classes occupy overlapping regions across much of the embedding space, although several localized clusters exhibit class predominance. The increased overlap suggests that the integration of CT-derived features introduces additional variability and feature complexity, resulting in a more heterogeneous latent structure. Despite this overlap, meaningful class-related organization is still apparent, indicating that multimodal radiomic features retain predictive information relevant to the target outcome.

A comparison between gray-level settings reveals that the overall embedding topology is preserved across both discretization schemes. However, the 128 gray-level PET/CT representation appears to provide slightly improved local grouping and reduced fragmentation of class-specific regions compared with the 64 gray-level configuration. This consistency supports the robustness of the extracted radiomic signatures to variations in intensity quantization.

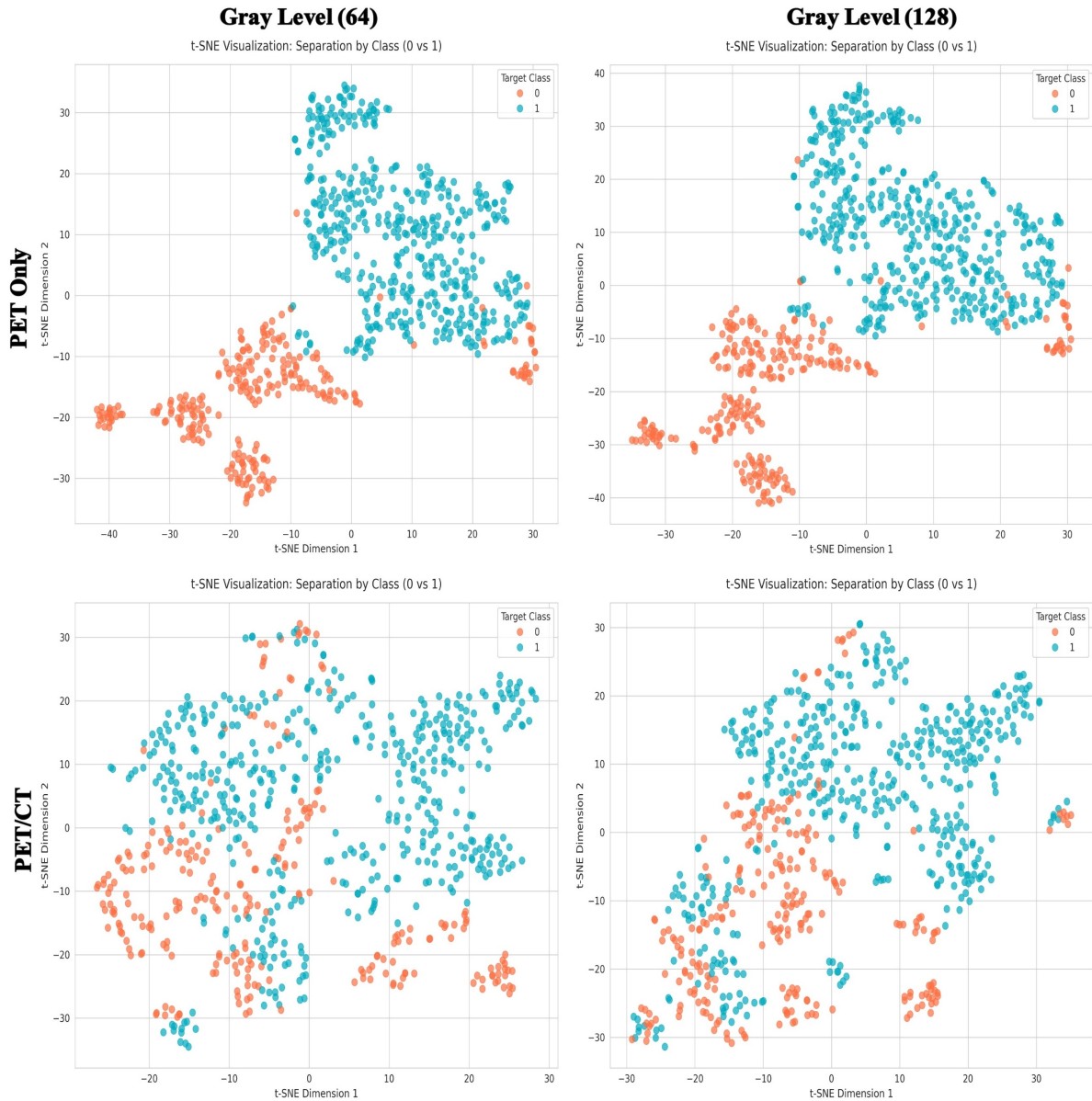


Figure 8. Two-dimensional t-distributed Stochastic Neighbor Embedding (t-SNE) visualization of lesion feature embeddings stratified by response class and gray-level quantization settings.

Figure 9 presents SHAP (SHapley Additive exPlanations) summary plots for the best-performing ML models identified under each imaging modality and gray-level discretization setting. The plots illustrate the contribution of the most influential radiomic features to model predictions, where the horizontal axis represents the SHAP value (feature impact on model output) and the color scale indicates the corresponding feature magnitude.

For the PET-only feature set at 64 gray levels (top-left panel), the Random Forest model achieved the highest mean AUC. The most influential feature was *GLSZM Large Zone High Gray Level Emphasis*, followed by a morphology-based shape descriptor incorporating radius sphere normalization and maximum intensity–perimeter relationships. Additional contributions were observed from *Intensity*

Skewness and *GLSZM Small Zone Emphasis*. The SHAP distributions indicate that lower values of the dominant texture feature were generally associated with increased positive model outputs, whereas higher values contributed negatively, suggesting that lesion texture heterogeneity plays a major role in prediction.

For the PET-only feature set at 128 gray levels (top-right panel), the Random Forest model again emerged as the optimal classifier, achieving a high level of predictive agreement ($\kappa = 0.922$). The most influential feature was *Intensity Histogram Maximum Gradient*, followed by *GLRLM Short Run Low Gray Level Emphasis*, *GLCM Inverse Variance*, and *Intensity Skewness*. Compared with the 64 gray-level model, the feature ranking shifted toward intensity-gradient and run-length texture descriptors, indicating that finer gray-level discretization enhances the relevance of subtle intensity-transition patterns. The broader SHAP distributions observed for these features suggest stronger discriminatory influence on model predictions.

For the combined PET/CT feature set at 64 gray levels (bottom-left panel), the best-performing model was XGBoost ($\kappa = 0.893$). The dominant predictor was *GLSZM Large Zone Emphasis* derived from CT imaging, followed by *Total Lesion FAPI (TLF)*, a morphology-based FAPI PET shape descriptor, and PET intensity histogram metrics. The substantially wider SHAP range compared with the PET-only models indicates stronger feature contributions and greater model sensitivity to variations in lesion characteristics. The prominence of CT-derived texture features suggests that anatomical heterogeneity contributes meaningfully to prediction when multimodal features are incorporated.

For the combined PET/CT feature set at 128 gray levels (bottom-right panel), XGBoost remained the optimal model, although with a lower agreement score ($\kappa = 0.769$). The most influential feature was again *GLSZM Large Zone Emphasis* from CT, followed by morphology-derived shape descriptors, *Minimum Intensity*, *NGTDM Busyness*, and *GLRLM Gray Level Non-Uniformity*. The persistence of CT texture features among the highest-ranked predictors across discretization schemes demonstrates their robustness and importance for classification. However, the increased concentration of SHAP values around zero for some lower-ranked features suggests a reduction in their marginal contribution under the higher gray-level setting.

Across all analyses, texture-based radiomic descriptors consistently emerged as the most influential predictors, particularly features derived from the GLSZM, GLRLM, and GLCM families. Morphological descriptors and intensity-based statistics provided complementary predictive information but generally exhibited lower overall influence. Furthermore, the stability of several top-ranked features across gray-level discretization settings indicates that the identified radiomic biomarkers are relatively robust to intensity quantization choices. Collectively, these findings demonstrate that both intralesional texture heterogeneity and lesion morphology contribute substantially to model decision-making, with PET-only models relying predominantly on metabolic texture characteristics and PET/CT models deriving additional predictive value from CT-based structural information.

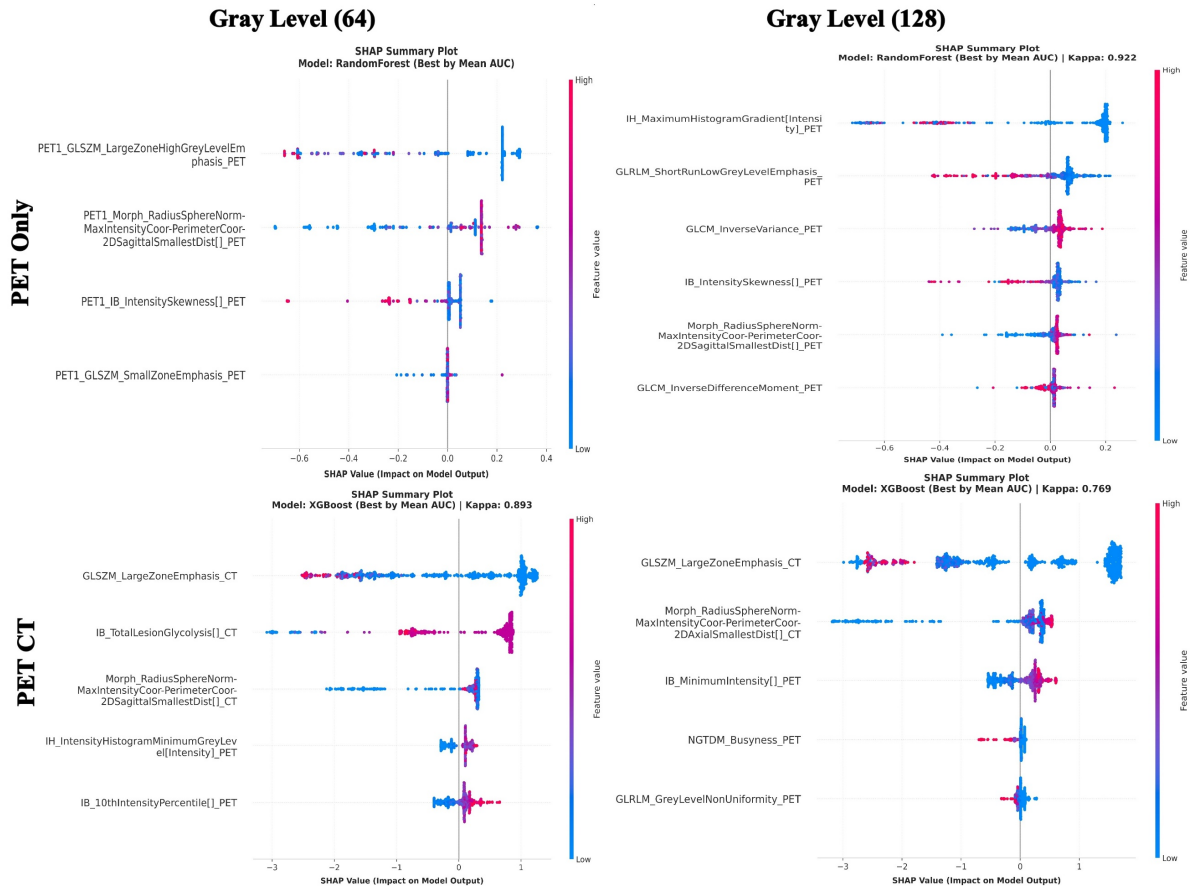


Figure 9. SHapley Additive exPlanations (SHAP)-based interpretation of the most influential radiomic features across gray-level discretization settings and imaging modalities.